

Structure-first mining of the metagenomic dark proteome finds serine hydrolases but does not extend PET-hydrolase discovery beyond sequence homology

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Abstract

Protein structure prediction and structural search are increasingly proposed as a route to discover divergent enzymes hidden in the metagenomic “dark proteome” — sequences beyond the reach of homology-based search — with poly(ethylene terephthalate) (PET) hydrolases a flagship target. Whether structure-first search genuinely extends *functional* discovery past the boundary of sequence homology has not been tested directly. We built a calibrated structure-first pipeline — fold-class structural search, a catalytic Ser–His–Asp geometry gate, and a size-invariant active-site-exposure score — validated to cleanly separate known PET hydrolases from non-PET serine hydrolases on crystal structures, and applied it to the ESM Metagenomic Atlas (~37M predicted structures), with a properly sampled random baseline and a per-anchor decomposition. Fold-class search enriches serine hydrolases ~53-fold over random, but the active-site-exposure score is not selective: it admits ~43% of random triad-bearing hydrolases. The PET-hydrolase branch is statistically indistinguishable from the random baseline at discovery distances, and the only above-baseline signal is confined to near-homolog sequence identities, where homology search already reaches. A maximum-sensitivity re-search retrieves more than nine million divergent (<20% identity) structural neighbors, yet these are *depleted* in the exposed-cleft phenotype relative to the random baseline, and molecular docking does not discriminate PET hydrolases from negatives. We conclude that structure-first triage is an effective serine-hydrolase finder but does not extend PET-hydrolase discovery beyond sequence homology’s reach; structure and sequence fail together in the divergent tail. We release the full pipeline and intermediate data as a reproducible benchmark.

Author summary

Enzymes that break down PET plastic are valuable, and there is growing hope of discovering new ones hidden in the vast, largely uncharacterized protein space of environmental metagenomes — regions that ordinary sequence-similarity searches miss. Recent advances in protein structure prediction have made it possible to search by predicted three-dimensional shape rather than by sequence, and this has been proposed as a way to reach those “dark” enzymes. We tested that premise rigorously for PET hydrolases. We built a structure-based screening pipeline, confirmed it cleanly recognizes known PET hydrolases among look-alike enzymes, and then ran it across roughly 37 million predicted metagenomic structures. The pipeline reliably finds the broad family of serine hydrolases, but the structural feature we used to flag PET-hydrolase-like active sites does not actually single them out — it passes nearly half of unrelated hydrolases — and the only useful signal sits at sequence distances a standard homology search already covers. Even when structural search pulls in millions of highly divergent candidates, they are if anything *less* likely to show the

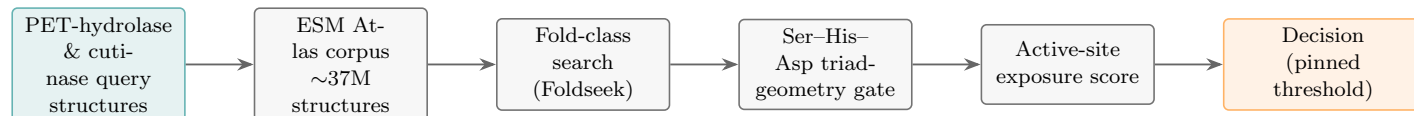


Figure 1: Structure-first pipeline. Reference PET-hydrolase/cutinase structures are searched by fold against the ESM Metagenomic Atlas; hits passing a catalytic triad-geometry gate are scored for active-site exposure and thresholded at a pinned, jitter-immune decision boundary. *Schematic; regenerate as author-owned artwork for submission.*

relevant active-site feature than random. Our results are a cautionary, fully reproducible benchmark: for this problem, predicted structure does not see further than sequence.

1 Introduction

Poly(ethylene terephthalate) (PET) is among the most abundant manufactured plastics, and enzymatic depolymerization has emerged as a promising route to its recycling and remediation. The discovery of *Ideonella sakaiensis* PETase and the engineering of leaf-branch compost cutinase (LCC) established that microbial serine hydrolases can hydrolyze PET, and motivated a broad search for additional and more capable enzymes [REF: Yoshida 2016; REF: Tournier 2020]. Environmental metagenomes are an attractive hunting ground: they encode enormous, largely uncharacterized enzymatic diversity, much of it in the so-called dark proteome — proteins too divergent in sequence to be annotated reliably by homology-based methods [REF: dark proteome].

The advent of fast, accurate protein structure prediction has reframed this search. ESMFold and the resulting ESM Metagenomic Atlas place hundreds of millions of predicted metagenomic structures within reach, and structural-search tools such as Foldseek enable querying this space by fold rather than by sequence [REF: Lin 2023; REF: van Kempen 2024]. Because structure is more conserved than sequence, structure-first search is widely expected to recover functionally relevant proteins that sequence search cannot — and several recent efforts have reported candidate PET hydrolases mined from metagenomic or structural databases on this basis [REF: Acosta 2025; REF: structure-mining preprint].

The premise underlying these efforts — that searching by predicted structure extends *functional* discovery beyond the reach of sequence homology — has, however, rarely been tested head-on. A structural search can retrieve a remote homolog, but retrieval is not discrimination: to discover a *novel* PET hydrolase one must distinguish PET-hydrolyzing geometry from the far larger background of serine hydrolases that share the α/β -hydrolase fold but act on other substrates. Here we ask, directly and at scale, whether an in-silico structure-first pipeline can do so. We assemble a calibrated pipeline, validate it on crystal-structure controls, apply it to the full ESM Metagenomic Atlas, and decompose its behavior against a properly sampled random baseline. We find that the pipeline is an excellent serine-hydrolase finder but that its PET-specific signal is bounded by sequence homology, and that in the divergent dark tail structure and sequence fail together.

2 Results

2.1 A calibrated structure-first pipeline

The pipeline (Fig 1) proceeds from a query set of PET-hydrolase and cutinase reference structures through (i) fold-class structural search against a target corpus, (ii) a catalytic Ser-His-Asp triad-geometry gate, and (iii) a cleft-detection and active-site-exposure scoring step that yields a composite decision score. We calibrated the pipeline on crystal structures: PET hydrolases (*I. sakaiensis* PETase, PDB 6EQE; LCC, 4EB0) as positives, and a panel of non-PET serine hydrolases (*Candida antarctica* lipase B, 1TCA; acetylcholinesterase, 1EA5; *Candida rugosa* lipase, 1CRL; *Alicyclobacillus acidocaldarius* esterase Est2, 1EVQ), together with a catalytically inactivated LCC variant (6THS) as a trap, as negatives. Under the calibrated threshold the positives separated from the negatives with a decision margin of +1.31 (raw) and +1.41 after the size-invariance

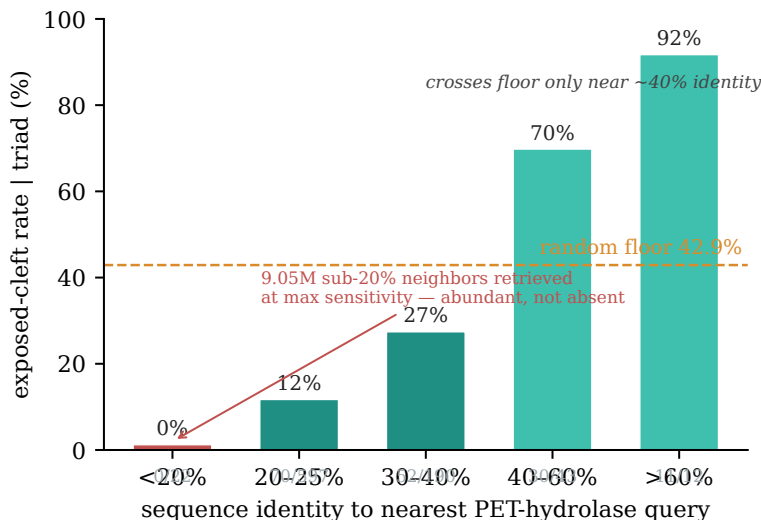


Figure 2: The PET-relevant signal is a homology gradient. Exposed-cleft rate among triad-bearing PET-hydrolase-branch hits versus sequence identity to the nearest query. The rate is monotonic and crosses the random baseline (dashed) only near $\sim 40\%$ identity; the $<20\%$ band is abundantly populated (9.05×10^6 retrieved neighbors) yet screens at zero. Bin counts shown below bars.

correction below, with no negative scoring above any positive (precision 1.0 on the control set). Because the cleft-detection step (fpocket) is non-deterministic between runs, we pinned the decision threshold (-1.1587 on the at-scale normalization) so that all metagenomic comparisons are immune to run-to-run jitter, and we normalized the exposure score to a size-invariant percentile so that it does not co-vary with protein or pocket size.

2.2 Fold-class search enriches hydrolases, but the cleft score is not selective

To separate what the pipeline’s stages each contribute, we drew a random baseline of 1,500 Atlas proteins (uniform byte-offset sampling, fixed seed) and passed them through the unchanged pipeline. The fold-class/triad stage is highly enriching: 99.3% of fold-class hits carry a catalytic triad versus 1.87% of random Atlas proteins, a 53-fold rate ratio (Fig 3). The exposure score, however, is not selective. Among random triad-bearing proteins, 42.9% (12/28) clear the exposed-cleft threshold (Wilson 95% CI 26.5–60.9%); that is, nearly half of unrelated serine hydrolases pass the very step intended to flag PET-hydrolase-like active sites. The precision observed on the four curated negatives — which happen to be buried- or gorge-active-site enzymes — does not generalize, because exposed active sites are common across the hydrolase background. Overall, only 0.80% (12/1500; Wilson CI 0.46–1.39%) of random proteins clear the full pipeline, so the pipeline’s enrichment is real, but it is carried entirely by the fold/triad stage, not by the exposure score.

2.3 At Atlas scale, the PET-hydrolase branch is flat against random

We searched the query set against the ESM Metagenomic Atlas high-quality clustered set (36,986,627 predicted structures, pLDDT > 0.7), yielding 217,833 fold-class hits (0.59% of the corpus). Ranking hits by global structural bit-score conflates query size with match quality (the larger acetylcholinesterase fold accrues higher absolute bit-scores), so we re-tiered hits *within each query anchor*. The PET-hydrolase/cutinase branch is in fact the largest, comprising 169,487 hits (77.8%), versus 41,697 (19.1%) for other-negative anchors and 6,649 (3.1%) for the acetylcholinesterase anchor. Screening the top 300 hits per anchor, the PET-hydrolase branch clears the exposure line at 32.65% (96/294 triad-bearers; CI 27.6–38.2%) — statistically indistinguishable from the 42.9% random baseline (Fisher $p = 0.30$, rate ratio 0.76). The acetylcholinesterase branch is lower (8.75%)

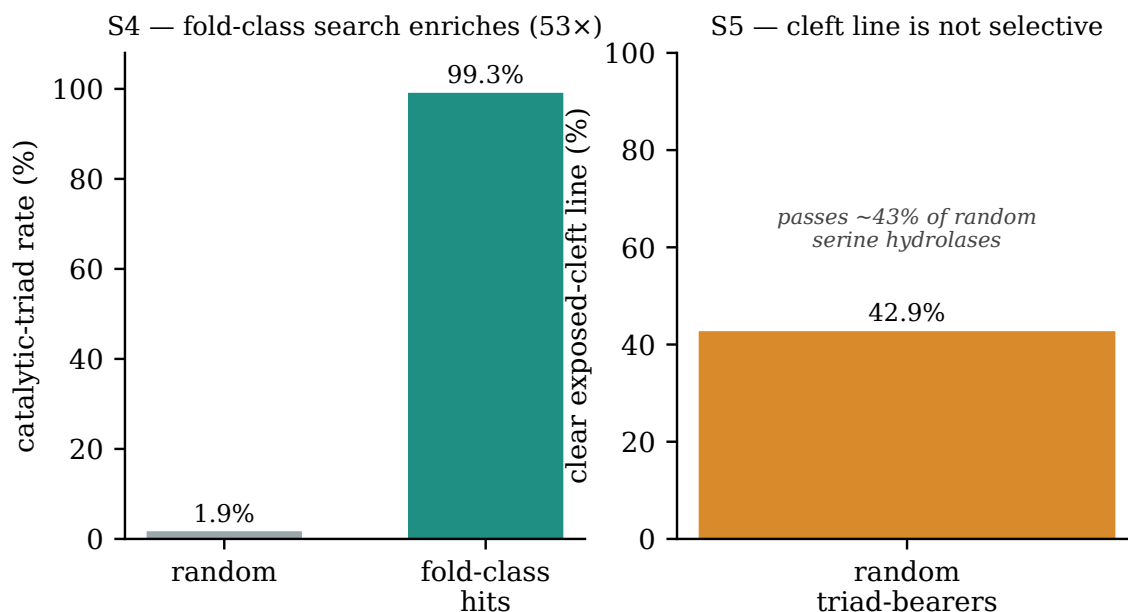


Figure 3: The lift is carried by fold-class search, not the cleft score. (Left) Catalytic-triad rate among fold-class hits versus random Atlas proteins (53-fold). (Right) Among random triad-bearing hydrolases, the fraction clearing the exposed-cleft threshold; nearly half pass, so the score is not selective.

only because its deep-gorge active sites fall *below* the baseline, not because the PET-hydrolase branch rises above it (Fig 4). Within the PET-hydrolase branch a genuine gradient exists, but it is confined to the closest neighbors (top-25 by bits, 80%; decaying to the baseline by ~ 1090 bits), foreshadowing a homology-bounded signal.

2.4 The PET-relevant signal is confined to near-homolog distances

To test the homology boundary directly, we recomputed the exposed-cleft rate of PET-hydrolase-branch triad-bearers as a function of *sequence identity* to the nearest query, rather than structural bit-score (Fig 2). The rate is monotonic in identity and crosses the random baseline only near $\sim 40\%$ identity: 91.7% above 60% identity, 69.8% at 40–60%, but 27.4% at 30–40% and collapsing below the baseline through the twilight zone (~ 25 –30% identity). Structural bit-score and sequence identity are strongly correlated here (Pearson $r = 0.735$), and the previously observed ~ 1090 -bit crossover maps to $\sim 35\%$ identity, above the twilight threshold; the structural and sequence gradients are one and the same. Triaging the 96 above-line hits, 93 lie at $\geq 30\%$ identity (near-homologs that sequence search already reaches) and none fall below 25% identity under the primary metric (one borderline case at 23.4% under the strictest structural identity measure, a full-coverage homolog of a known archaeal PET hydrolase, not a novel divergent fold). Sequence-identity estimates were cross-validated against Foldseek’s own alignment identity ($r = 0.987$).

2.5 The divergent tail is abundantly reachable but depleted

A natural objection is that the empty divergent tail reflects a search-sensitivity limit rather than biology. It does not. A maximum-sensitivity re-search (Foldseek `-s 9.5`, near-zero bit-score floor) retrieves 9,810,546 unique targets, of which 9,050,775 lie below 20% identity to any query — a more than fifty-fold increase over the default search and a clearly abundant divergent tail. We screened 2,100 of the most structurally convincing sub-25%-identity hits, stratified across identity bands. Triads are common among them (1,710/2,100), but only 11.3% of credible-coverage sub-25% triad-bearers clear the exposure line — far below the 42.9% random

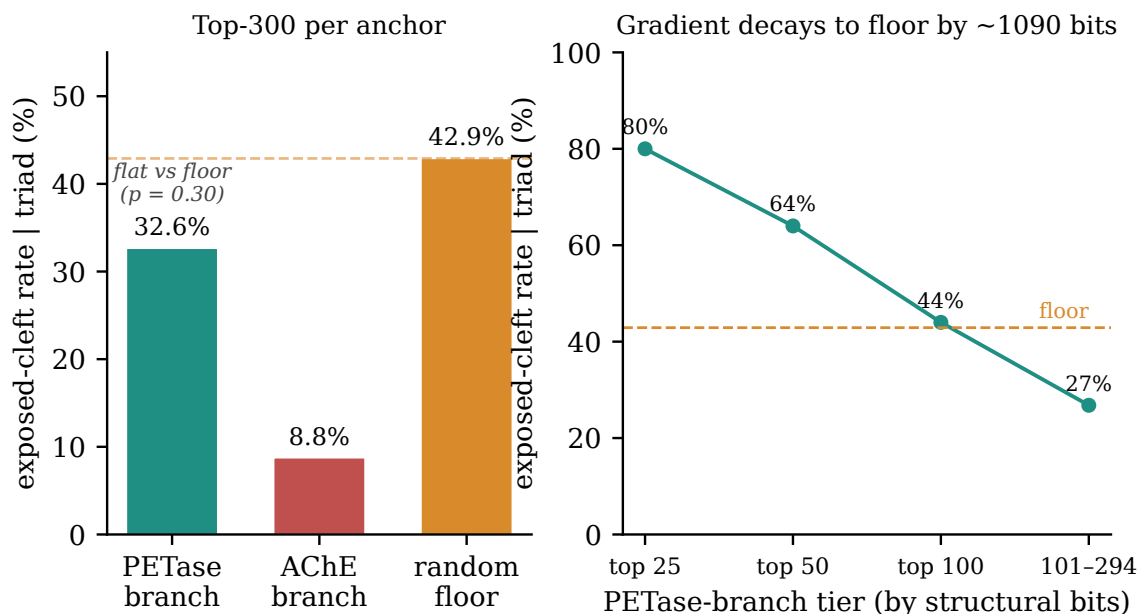


Figure 4: The PET-hydrolase branch is flat against the random baseline. (Left) Exposed-cleft rate for the top 300 hits per anchor: the PET-hydrolase branch is statistically indistinguishable from the random baseline (dashed; $p = 0.30$), and the acetylcholinesterase branch sits below it. (Right) Within the PET-hydrolase branch the gradient decays to the baseline by ~ 1090 structural bits.

baseline (rate ratio 0.27, $p < 10^{-4}$) — and none of the 22 screened below 20% identity clear it. (Forty-seven above-line hits with low alignment coverage, median 7%, were flagged as coverage artifacts and excluded; they do not change the result.) Thus structure reaches the divergent tail in abundance, but the exposed-cleft phenotype is, if anything, depleted there relative to the random hydrolase background: retrieval is not the limitation, discrimination is.

2.6 Docking does not discriminate

As an orthogonal specificity check, we docked a PET fragment analog (bis(2-hydroxyethyl) terephthalate, BHET) into positives and negatives. Predicted binding affinities overlap across PET hydrolases and non-PET hydrolases within the scoring function’s error, providing no separation beyond the structural pipeline. Docking is therefore confirmatory of, not a remedy for, the absence of an in-silico specificity signal.

3 Discussion

Across calibration, an Atlas-scale screen, a random-baseline decomposition, a sequence-identity reach analysis, and a maximum-sensitivity re-search, a consistent picture emerges. Structure-first triage is an effective *finder* of serine hydrolases — the fold/triad stage enriches them 53-fold — but it is not a *discriminator* of PET-hydrolyzing function. The active-site-exposure score, which cleanly separated PET hydrolases from a curated panel of buried-site negatives, admits roughly 43% of random triad-bearing hydrolases and provides no separation for the PET-hydrolase branch against the random baseline. Whatever PET-relevant signal exists is confined to near-homolog sequence identities, precisely the regime a standard homology search already covers, and it vanishes by the twilight zone. Where structure should add value — the divergent dark tail — it adds none: the tail is abundantly reachable but depleted in the relevant phenotype, so structure and sequence fail together rather than complementing one another.

The depletion of the exposed-cleft phenotype in the divergent tail is a robust observation, and we state it

as such: the exposed-cleft rate declines monotonically with sequence divergence and falls below the random-hydrolase baseline among the most divergent neighbors. We are more cautious about *why*. The comparison is between two different populations — a random draw of hydrolases versus sequence-divergent structural neighbors of PET hydrolases — and so the depletion should not be over-interpreted. One plausible account, which we offer as a hypothesis rather than a conclusion, is that structural search at low sequence identity matches the conserved α/β -hydrolase fold core, while the surface-exposed, solvent-accessible cleft that PET hydrolases require for attack on an insoluble polymer is a peripheral, specialized adaptation that is rare among divergent members of the fold.

These results are complementary to, not in conflict with, reports of metagenome-mined PET hydrolases. Such efforts have typically validated candidates that are themselves near-homologs of known enzymes, or have relied on homology-seeded entry filters; our analysis quantifies the boundary within which those pipelines operate, rather than contradicting their experimentally confirmed hits. The practical implication is a cautionary one: in-silico structure-first search should not be expected, on its own, to deliver *novel* PET hydrolases from the divergent dark proteome, and claims of structural “discovery” beyond homology warrant explicit demonstration against a random baseline and against a sequence-identity reach analysis of the kind reported here.

Several limitations bound our conclusions. The Atlas structures are ESMFold predictions; we mitigated model-quality confounds by restricting to the high-confidence clustered set (pLDDT > 0.7) and by coverage-gating sequence-identity estimates, but prediction error at low identity cannot be excluded entirely and merits an explicit pLDDT comparison between the divergent band and the baseline. Our specificity signals — active-site exposure and BHET docking — are particular operationalizations; a richer, PET-specific descriptor (for example, an aromatic substrate-binding subsite) might in principle perform differently, although the same low-identity model noise that limits exposure scoring would bear most heavily on such a descriptor, and our docking result is not encouraging. Finally, sequence identity here measures the reach of sequence search, not PET activity itself; all candidates are computational priorities, not experimentally validated enzymes, and wet-lab characterization remains necessary to establish function.

4 Materials and methods

Controls and calibration. Positive controls: *I. sakaiensis* PETase (PDB 6EQE) and LCC (4EB0). Negative controls: CalB (1TCA), acetylcholinesterase (1EA5), *C. rugosa* lipase (1CRL), Est2 (1EVQ), and a catalytically inactivated LCC variant (6THS). The catalytic triad was defined geometrically on Ser–His–Asp side-chain atoms; the composite decision score combines triad geometry with an active-site-exposure (peripherality) term derived from fpocket pocket detection, normalized to a size-invariant percentile. The decision threshold was pinned at -1.1587 on the at-scale normalization to ensure invariance to fpocket’s run-to-run non-determinism.

Corpus and structural search. The target corpus was the ESM Metagenomic Atlas high-quality clustered set (36,986,627 structures, pLDDT > 0.7), queried with a prebuilt Foldseek database. Default fold-class search used standard Foldseek sensitivity; the divergent-tail re-search used `foldseek -s 9.5 --max-seqs 4000000 -e 10000` with a near-zero bit-score floor. Hits were tiered within each query anchor by anchor-specific bit-score. Structures for screening were fetched on demand (Foldcomp / predicted-structure API).

Baseline sampling. The random baseline was 1,500 Atlas proteins drawn by seeded (1729) uniform HTTP byte-range offset sampling and passed through the unchanged pipeline at the pinned threshold.

Sequence identity. Pairwise identity to the nearest query was computed by Smith–Waterman alignment (biotite), reported with alignment coverage and gated for low-coverage artifacts, and cross-validated against Foldseek alignment identity ($r = 0.987$).

Docking. BHET was docked into control structures with AutoDock Vina; affinities were compared across positives and negatives.

Statistics. Proportions are reported with Wilson 95% confidence intervals; group comparisons use Fisher’s

exact test with rate ratios. fpocket non-determinism was controlled by threshold pinning and matched re-screening.

Compute. Local triage on Apple Silicon; burst structural search on a single cloud instance (16 vCPU), with the divergent re-search completing in ~13 minutes.

Data and code availability

Pipeline source, configuration, and analysis scripts: github.com/akarlin3/projProteus [confirm public release commit/tag]. Intermediate results, per-hit tables, and run reports are archived in cloud storage and will be deposited to a permanent repository (e.g. Zenodo) with a citable DOI prior to submission [to mint]. The corpus is the public ESM Metagenomic Atlas.

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